

Class 17 Q13-14

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Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes. Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the medium value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `medium()` and `summary()` function that you can use to check your understanding.

```
file_path <- "sample geno exp.txt"

my_table <- read.table(
  file = file_path,
  header = TRUE,
  sep = " ",
  skip = 0
)
```

```
table(my_table$geno)
```

```
A/A A/G G/G
108 233 121
```

```
tapply(my_table$exp, my_table$geno, median)
```

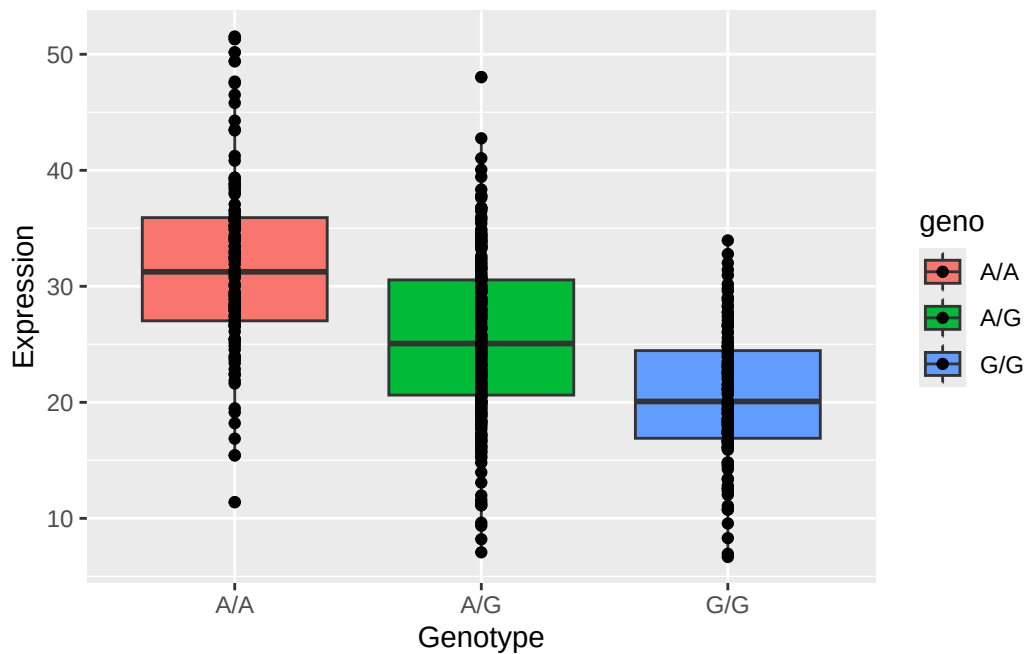
```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
ggplot(my_table, aes(x = geno, y = exp, fill = geno)) +  
  geom_boxplot() +  
  labs(x = "Genotype", y = "Expression") +  
  geom_point()
```



G/G has a lower expression value compared to A/A and A/G at this position/locus, and the SNP does affect the expression of ORMDL3 in that a copy of the G allele will reduce the expression of this gene. This explains why G/G has the lowest expression value and why A/A has the highest expression.